

Mandi to Market Optimizer

A comprehensive solution integrating data, analytics, and decision intelligence for the agricultural supply chain.

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The Problem

India's agricultural supply chain generates **data but lacks intelligence**, creating challenges for effective decision-making.

- Messy crop names
- Mixed units
- Missing values
- Invalid transit times
- Different timestamps

Our Solution

Mandi Intelligence

Transforming messy datasets into actionable insights requires a **streamlined process** for effective decision-making and analysis.

**RAW DATA → CLEANING → SQLITE
→ ANALYTICS → DASHBOARD → AI AGENT**

Data Trustworthiness

Standardization

Implementing consistent **crop name** and Mandi ID formats ensures a reliable data foundation for analysis.

Conversion

Converting measurements like quantity to quintals and distance to kilometers enhances data integrity and usability.

Validation

Identifying and excluding invalid records, such as negative transit times, guarantees data quality throughout the pipeline.

57 Mandis 25,000 Arrivals 12,000 Prices 10,000 Trips 15,000 Weather Records

Analytics Engine

5,659,184.8
2 Qtl

Total crop arrivals

3,924

Price crash instances

18.68%

Transport delay rate

342

Routes Monitored

01

Decision Signals

This analytical layer produces insights for decision-makers, providing multiple signals from cleaned datasets for actionable intelligence.

02

SQLite Database

Utilizing SQLite, the architecture efficiently stores and exposes various analytical views, enhancing data retrieval for stakeholders.

03

Clean Architecture

A streamlined structure supports data flow, ensuring that cleaned data effectively feeds into the analytics layer for reliable analysis.

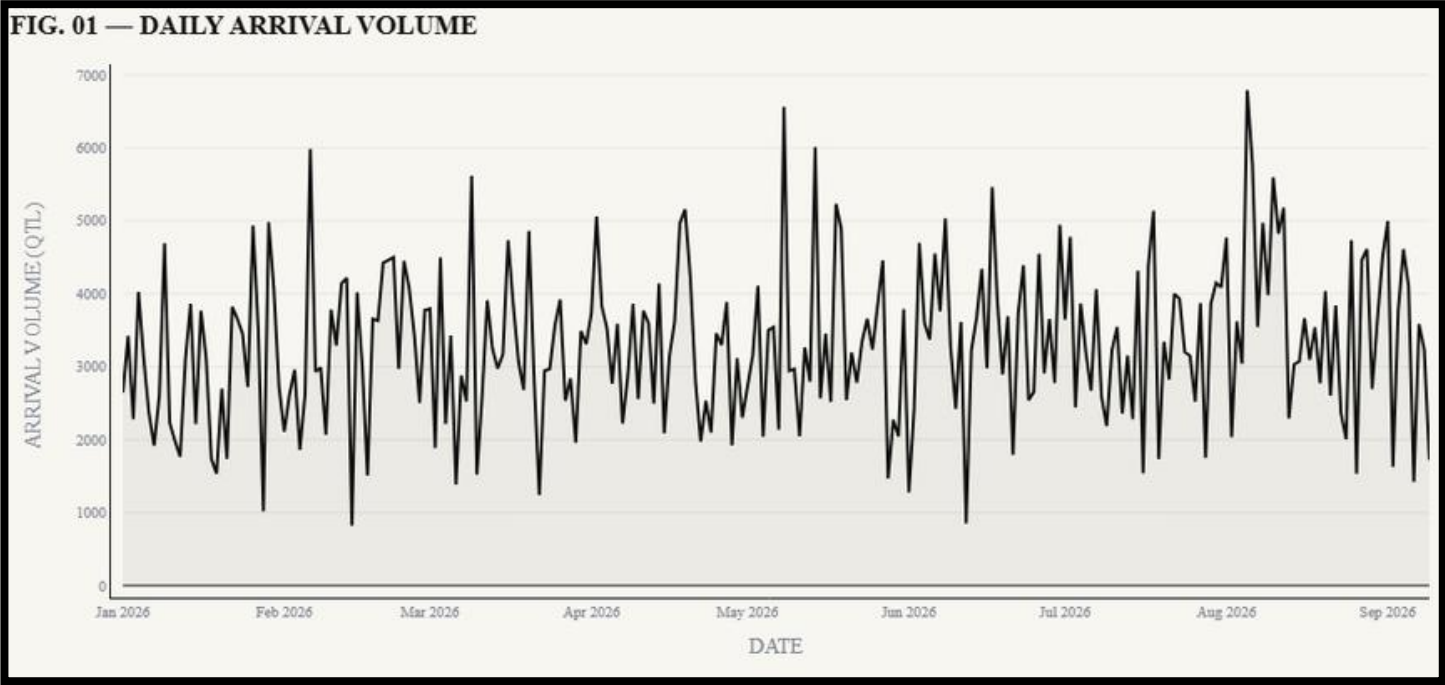
TOP 5 MANDIS BY ARRIVAL VOLUME (QTL)

- 1. South Dumdum Grain Market — 20,539
- 2. Arrah Mandi — 20,454
- 3. Gurgaon Grain Market — 18,923
- 4. Asansol Grain Market — 17,948
- 5. Hyderabad Mandi — 17,863

CROP-WISE ARRIVAL DISTRIBUTION (QTL)

Mustard 966,396 Wheat 965,396 Sugarcane 949,853
Maize 940,987 Cotton 922,311 Rice 914,242

Supply Intelligence



Source: AgriTech Analytics Database

Ranked by arrival volume and crop-wise distribution, drawn from the cleaned Crop Arrivals dataset. The daily trend excludes the dataset's blank-date aggregate row (4,825,951.80 Qtl).

Price Discovery

When wholesale price falls below MSP

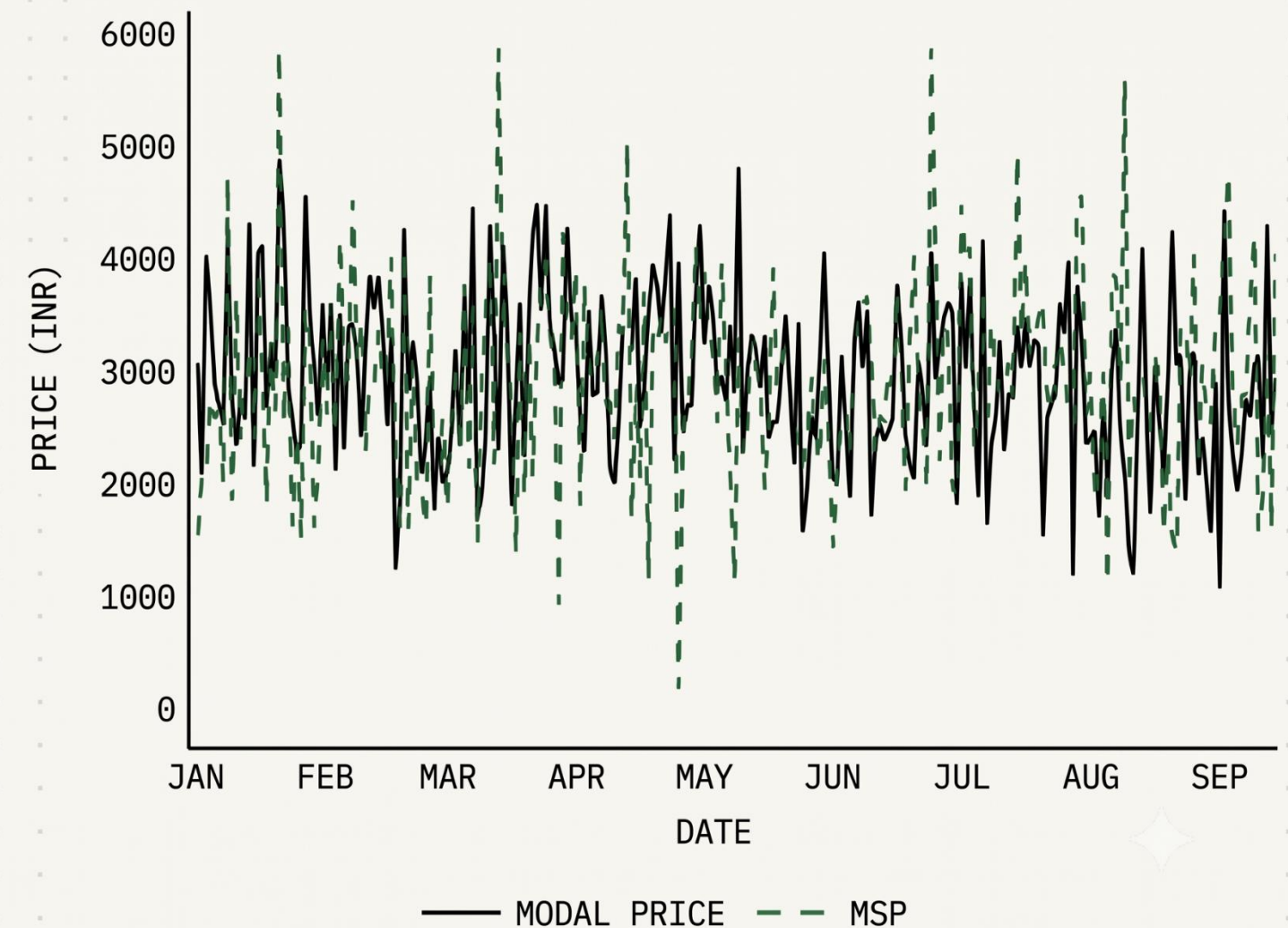
3,924

Price Crash Instances

Definition: Price crash = Modal Price < MSP. 610 records missing Modal Price. 2,377 records missing MSP. Does not establish why crashes occurred.

PRICE DISCOVERY

FIG. 03 — PRICE DISCOVERY (MODAL VS MSP)



Logistics Intelligence

Routes Monitored



342 monitored routes. 8,455 valid transit trips. 539 invalid negative transit records excluded from calculations.

Delay Methodology



A route is flagged delayed when transit time exceeds 1.5x the average valid transit time for that Mandi to Warehouse route.

Delay Rate



1,584 delayed trips. 18.68% transport delay rate. Identified delay patterns, not a claim of reduced delays.

Weather × Supply

CURRENT DASHBOARD VIEW

52 weather-matched arrival days

3,276 unmatched days retained with NULL weather metrics
(filtered view — not a global project total)

Observed Relationship

Rainfall vs arrival volume, aggregated to daily grain before joining with arrivals, preventing many-to-many inflation. Not a causal claim.

WEATHER × SUPPLY

Daily rainfall and temperature are compared with crop arrival volumes where weather observations are available.

**FIG. 04 —
OBSERVED
RELATIONSHIP:
RAINFALL ×
ARRIVALS**

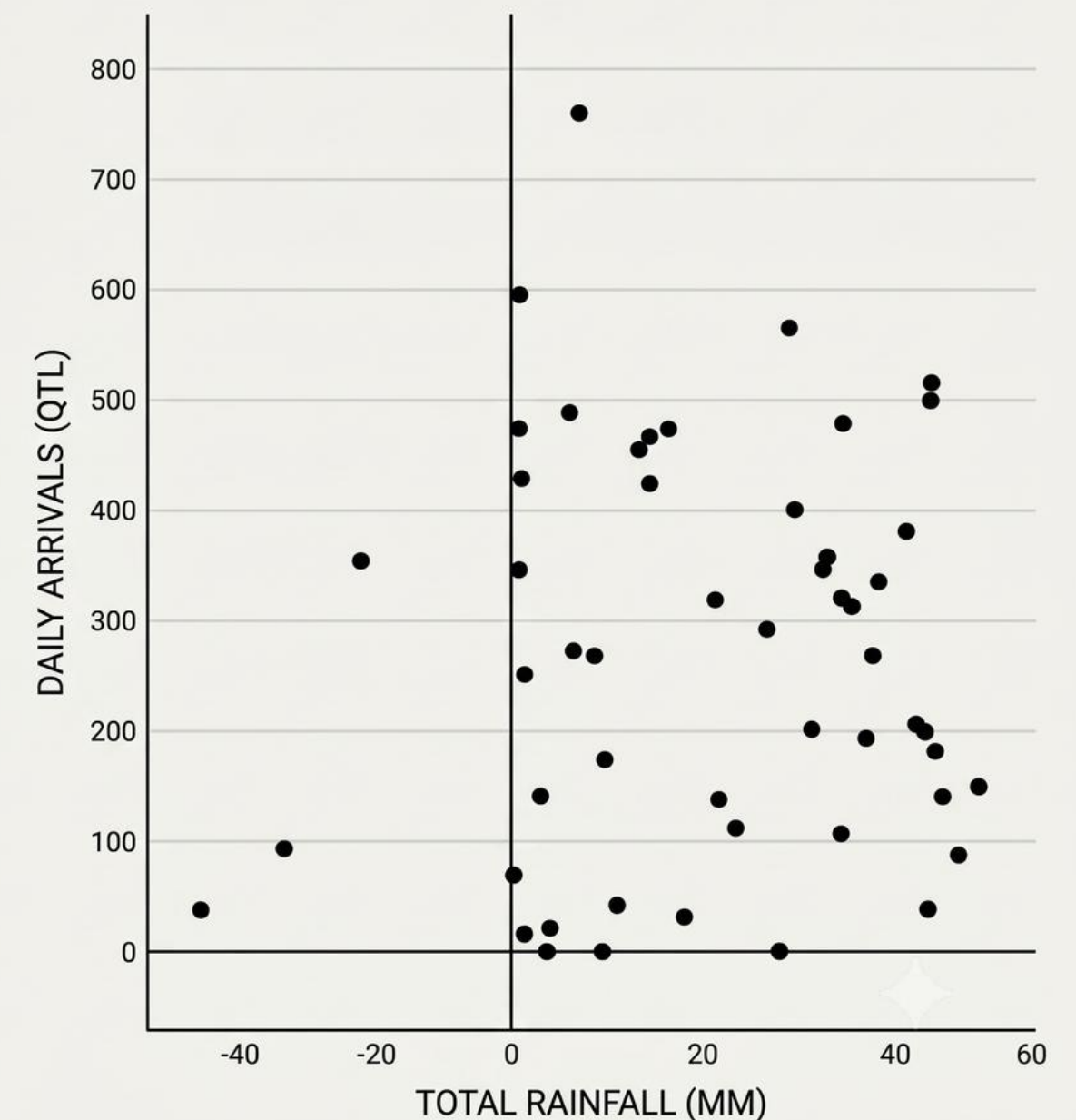
WEATHER COVERAGE

52

WEATHER-MATCHED
ARRIVAL DAYS

3,276

UNMATCHED DAYS
RETAINED

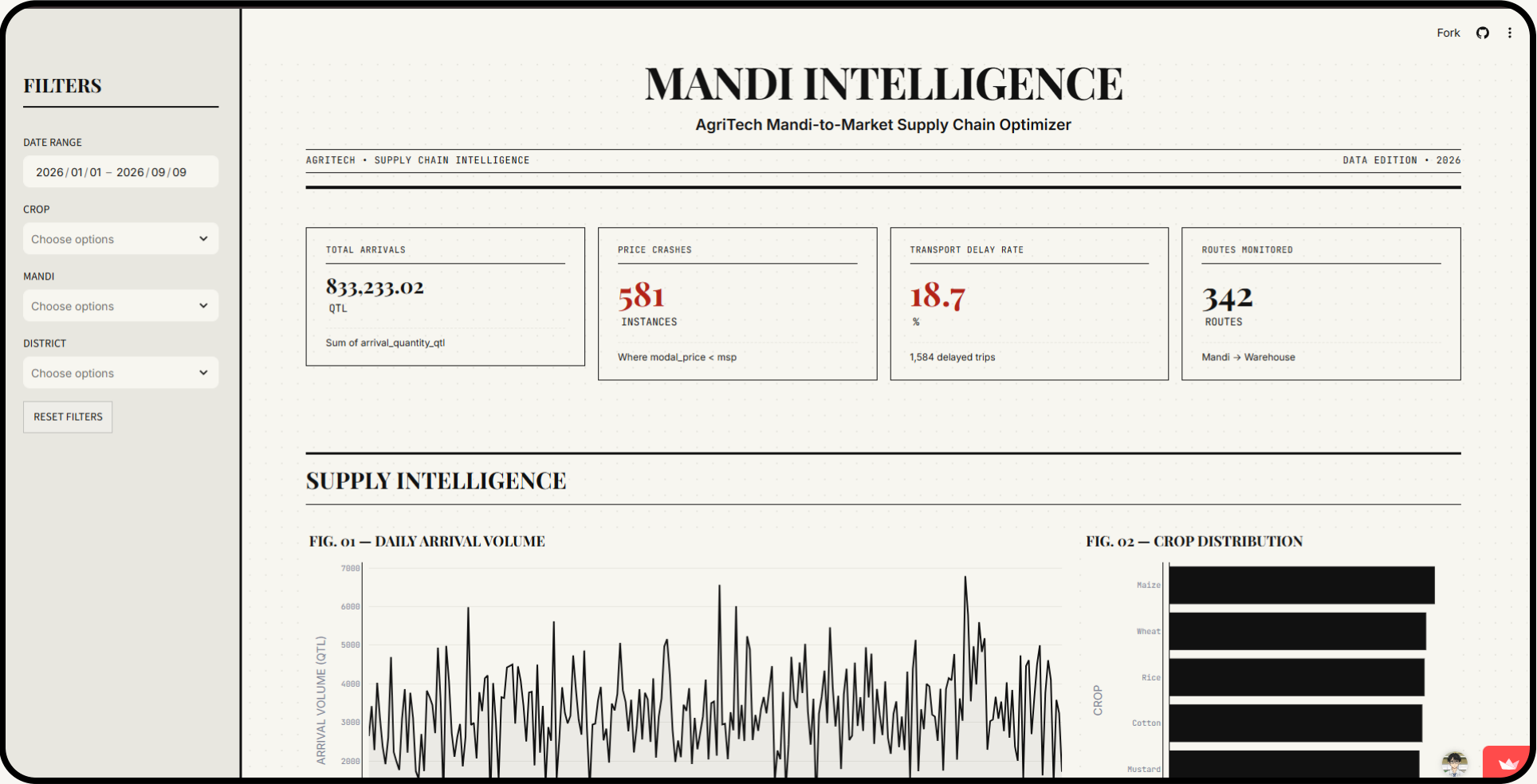


Executive Dashboard

One Screen for Mandi to Market Intelligence

This dashboard provides a comprehensive overview of critical agricultural data, integrating supply, pricing, logistics, and weather information for informed decision-making across the supply chain.

Interactive Streamlit dashboard for supply-chain intelligence



AI Query Agent

USER QUESTION → AI QUERY AGENT → ANALYTICAL QUERY → DATA-BACKED RESULT → CHART + TEXT SUMMARY

EXAMPLE QUESTIONS

Show Wheat arrivals in Amritsar mandi.

Which mandis have prices below MSP?

What is the highest average transit delay?

Show rainfall by district.

What are the highest-volume warehouses?

Ask the Data Instead of Searching Through Charts

Ask questions about the supply chain data in plain English. The agent converts your query into SQL and generates charts automatically.



Technical Architecture

Raw Datasets



Data Cleaning Pipeline



Clean Dataset



SQLite Analytics Database



Validated SQL Views



Streamlit Dashboard



AI Query Agent

01

Data Pipeline

A streamlined **data pipeline** efficiently connects raw datasets to a clean analytics database for effective decision-making.

02

Reproducibility

The architecture ensures **reproducibility**, allowing for consistent data cleaning and analytical processing through well-defined Python scripts.

03

Technology Stack

Built using Python, SQLite, and Streamlit, the **technology stack** supports robust data handling and visualization capabilities.

Validation & Reproducibility



DATA QA / VALIDATED

- ✓ Cleaned row counts reconciled
- ✓ Duplicate handling verified
- ✓ Arrival totals reconciled
- ✓ Weather join checked for many-to-many inflation
- ✓ Price crash condition manually validated
- ✓ Invalid transit records excluded from valid calculations
- ✓ Route delay methodology validated
- ✓ Analytical views tested against source data

Decision Intelligence

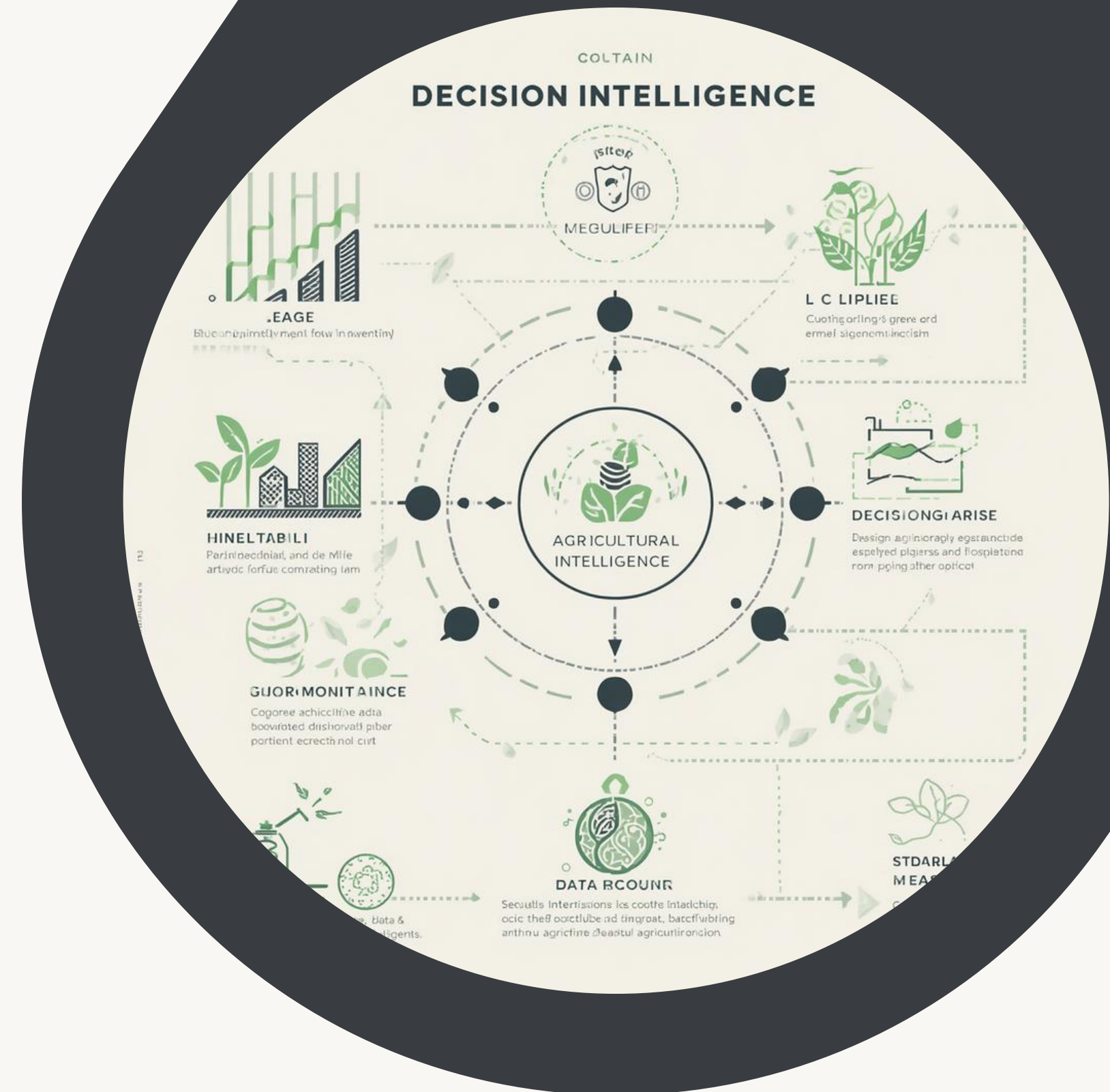
SUPPLY — Where are arrivals concentrated?

PRICE — Where is modal price below MSP?

LOGISTICS — Which routes show higher delay frequency?

WEATHER — Where do rainfall patterns coincide with supply changes?

MARKET SIGNALS — What requires attention?



Conclusion

AGRITECH INTELLIGENCE

AgriTech transforms fragmented agricultural data into trusted supply-chain intelligence.

Clean Data → Validated Analytics → Interactive Dashboard → AI-Powered Insights

MANDI → SUPPLY → PRICE → LOGISTICS → WEATHER
→ INTELLIGENCE

GitHub: github.com/httpsankuu/agriTech

Live Dashboard: agritech-byankith.streamlit.app

Python · Pandas · SQLite · SQL · Streamlit · Plotly ·
Custom CSS

